

ResearchPilot Sample Knowledge Base

Fictional internal document created for testing the RAG component of ResearchPilot.

1. Organization Overview

NovaTech AI Lab is a fictional software team that develops internal AI assistants for research, knowledge discovery, and operational support.

The team follows a simple principle: the model should answer from trusted evidence whenever evidence is available and should clearly state when required information is missing.

2. ResearchPilot

ResearchPilot is the lab's agentic research assistant. It can search approved live sources, search the private document collection, summarize evidence, and save research reports.

The project code is **NP-01**. Generated reports are stored in the project's **reports/** directory, while private source documents are stored in **documents/**.

3. System Architecture

ResearchPilot uses a stateful agent architecture. A user gives the system a goal. The language model decides whether additional information or an action is required. The agent can call one or more tools, observe their results, and continue until it has enough information to produce a final answer.

4. Internal Knowledge Retrieval

The internal knowledge base is built by loading documents, splitting them into meaningful chunks, creating vector embeddings, and storing those embeddings in a vector database.

The prototype uses **Chroma** as its vector store. The RAG search tool should return only the most relevant passages and their source metadata.

5. When to Use Each Source

Use **internal knowledge search** for private project details, internal policies, architecture notes, and company-specific information.

Use **web search** when the user asks for current, external, or time-sensitive information.

For complex requests, the agent may use both sources and should distinguish internal evidence from external evidence in its final answer.

6. Agent Tools

search_web(query) searches approved live web sources and returns concise results.

search_knowledge(query) searches the internal vector store and returns relevant passages.

save_report(title, content) saves a Markdown report inside the approved reports directory.

7. Multi-Step Research Procedure

Step 1: Interpret the user's goal and identify the required information or actions.

Step 2: Search internal knowledge when the request concerns private or project-specific information.

Step 3: Search approved live sources when current external information is required.

Step 4: Compare, organize, and summarize the evidence.

Step 5: Save a report when the user requests one.

Step 6: Return a concise response and report the actions completed.

8. Example Internal Facts

- Project code: **NP-01**
- Primary orchestration framework: **LangGraph**.
- Prototype vector store: **Chroma**.
- Primary model provider: **Google Gemini** through LangChain.
- Report format: **Markdown (.md)**.
- Approved report folder: **reports/**.
- Private document folder: **documents/**.

9. Safety Rules

- Tools must validate their input arguments before execution.
- The report-writing tool is restricted to the project reports directory.
- The agent must not be given unrestricted shell execution or arbitrary file deletion.
- Temporary service failures may be retried a limited number of times; persistent failures should be surfaced clearly instead of claiming success.

10. RAG Evaluation Questions

- What vector store does the prototype use?
- What is the ResearchPilot project code?
- Which framework orchestrates the agent?
- Where are generated reports stored?
- When should internal knowledge search be used?
- What safety restriction applies to the report-writing tool?

11. Test Scenario

A user asks: "Explain how ResearchPilot combines internal knowledge with live web research and save the explanation as a report." A correct execution should retrieve relevant internal evidence, use web search when current external information is required, summarize the evidence, save the report, and confirm completion.

Notice: This document is entirely fictional and exists only as a safe sample knowledge base for development and RAG testing.